

SOFT30121: Advanced Analysis and Design

Systems Analysis Design and Implementation (submission 1)

By

N0012345: Your Name

PROJECT TITLE HERE



SOFT30121: Advanced Analysis and Design1

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1. Introduction

Author(s):

Give an introduction to your project. Keep it high-level where you describe in-general what you have done/will do and for whom.

Purpose

The purpose of this project is to provide a standalone software system, which can be incorporated into Future Fridges' latest smart fridge model, the FFSmart. This software base; once integrated, will allow commercial restaurant businesses to have automated tracking of quantity and expiration of all products, a hands-free inventory replenishment system and auto-order capabilities.

Features integrated will allow for less manual input and automated notifications to keep users informed and reminded of the state of their inventory. Security will be ensured by incorporating user access level, which in turn will provide users with respective access to the system. Accounts will be protected using secure authentication to protect information and external access to the system.

The software system will be a user application which will present a simple and intuitive UI (User Interface), requiring minimal training for use. The load handling of simultaneous user usage should minimally affect performance, as well as maintaining good reliable response times from the load of user inputs.

In sense, this software system will stand as the bases for a new innovative framework for the future of fridges and hopefully grow into a standard amongst businesses within the commercial restaurant industry.

User Characteristics

The FFSmart system will be used by a variety of user groups involved in the operation and management of food storage within commercial restaurants. Users with the group have distinct responsibilities, access levels and technical skills. The system shall provide user interfaces and tailored featured to needs, to ensure security and ease of use.

Features and access:

Head Chef account: (moderate computer and mobile app skills)

- Full access to general overview of activity
- Quantity and expiration date notifications (Given a 3-day warning notification before items run out or are about to expire)
- Order confirmations and reorder confirmations
- Allow to configure chef account access to specific items in the inventory. (Split chefs into groups which will have specified food items they're eligible to take out)
- Health and safety documentation report attached
- Input quantity and what item has been removed from the fridge
- Allow to add additional members
- Allow to make amends to aspects of user profiles

Chef accounts: (Basic computer/app experience)

- Can log in to record added and removed items
- Allowed to view inventory

Delivery accounts: (Basic computer/app experience)

- Delivery account which will allow limited access for stock mark-off.
- Access to rear door
- Delivery Confirmation
- Login credentials to access the delivery entrance

System Administrator: (High technical expertise)

- Full system access, including user management, security settings, data backups, and internal restaurant data

Health and Safety Officer: (Basic digital literacy)

- Read only access to view or export system generated compliance reports

Assumptions

Hardware and Equipment:

It is assumed that the FFSmart fridge hardware is fully developed and includes all necessary sensors and communication modules.

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Assumes for stable connectivity between hardware and software for communication purposes.

All users will have access to a computing device that can run the software.

Software and System:

System will run on a standard operating system with appropriate runtime environment.

A relational database system will be available to store inventory, user data and supplier data.

Assumes an internet connection is consistently provided for data synchronization, notification sending and automated reordering.

Backend server has sufficient storage and processing capacity for handling concurrent user usage and data logging.

User Expertise and Training:

Assumed that all users have basic literacy and a moderate understanding of computing/application familiarity.

Administrative users and Head chef are assumed to have moderate technical knowledge.

Operational and Business:

All staff and delivery personnel must have a valid login credential and assigned access associated with roles.

Supplier information must be correctly registered into the database.

Deliveries will occur during defined business hours, and any data discrepancies will be handled manually.

All operations will comply with Health and Safety regulations.

Development and Testing:

Selected software development topology will be agreed upon with Future Fridge Ltd.

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Development team will have access to testing environments simulating fridge hardware.

The client will be available for regular feedback and validation during development and testing phases.

Scope and Constraints

The FFSmart software system is designed to support Future Fridge Ltd's solution for the commercial restaurant sector. The software will effectively monitor, manage and replenish food inventory stored within the FFSmart fridges.

This system will:

- Track food items being removed or added in real time
- Record user actions through secure logins
- Send automated alerts for items nearer expiration or in low quantity
- Generate Health and Safety reports
- Automate reordering from suppliers

The initial version will operate as a standalone simulation system independent of physical hardware. Future versions will be integrated into actual FFSmart fridges once everything is finalized.

Constraints:

- Must maintain audit trail of user activities, inventory updates, deliveries and automated reorders
- Must enforce role-based access control (RBAC) to ensure users are authorized to access features
- Data transfers between system and server must be encrypted
- Passwords shall be stored securely
- Must be developed using industry standard technologies and open-source frameworks where possible.

Application Criticality:

- Software is moderately critical, as it directly affects restaurants operations, food safety and supply management. Failure could lead to food spoilage, inventory loss or failure to comply with regulations.

System Constraints:

- Application should be optimized for use on mobile/tablet devices.
- Database storage should be scalable but constrained to avoid excessive resource consumption.
- Network communication should be able to handle low bandwidth

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locations.

Out of scope

Glossary of terms

User Interface (UI)

Role-Based Access Control (RBAC)

(Context based terms)

Overview

This document presents a complete break down and overview of functional and non-functional requirements for the project. While also including use case model and an overall extensive project plan to ensure complete understanding of the system before design and implementation.

2. Functional Requirements

Restaurants must be able to effectively manage their food inventory with the FrostByte system's intelligent automation, tracking, and notifications. The MoSCoW technique (Must, Should, Could, Won't) is used to prioritize functional needs and categorize them by feature.

Feature 1

User Authentication and Access Control (Must)

- FR1.1: Three user roles—head chef, delivery person, and food server—will be supported by the system.
- FR1.2: All users must provide their login information before they may access any features of the system.
- FR1.3: The system will limit access according to user roles: food server (no access), delivery person (only rear-door access), and head chef (complete access).
- FR1.4: The system will record, along with timestamps, each user's access session.

2.1. Feature 2

Inventory Management (Must)

- FR2.1: Every food item that is put into or taken out of the refrigerator must be recorded by the system.
- FR2.2: The item's name, amount, supplier, date of addition, and expiration date will all be tracked by the system.
- FR2.3: When items are scanned or logged, the system will instantly update inventory data.
- FR2.4: Duplicate entries for identical items with the same supplier and expiration date must be avoided by the system.

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2.2. Feature 3

Notifications and Alerts (Should)

- FR3.1: When an item is within three days of expiring, the system will notify the Head Chef's app.
- FR3.2: When an item's stock level is expected to run out within three days, the system will notify users.
- FR3.3: The item name, current quantity, and suggested reorder supplier must all be included in notifications.

2.3. Feature 4

Automated Reordering (Should)

- FR4.1: Every Monday, an automated reorder list will be generated by the system.
- FR4.2: Every item will be matched with its appropriate provider by the system.
- FR4.3: Prior to dispatch, reorders may be reviewed, confirmed, or canceled by the head chef.

2.4. Feature 5

Delivery Management (Must)

- FR5.1: Delivery staff will be able to view delivery schedules and designated refrigerators via the delivery app.
- FR5.2: Delivery staff will be able to access the back door using a protected login on the system.
- FR5.3: All restocked items must be entered into the system, which will then compare them to the delivery manifest.

2.5. Feature 6

Reporting (Could)

- FR6.1: The Health and Safety Officer will get compliance and hygiene reports from the system.
- FR6.2: Temperature logs, item histories, and expiry tracking must all be included in reports.

3. Non-Functional Requirements

These specify the general characteristics and limitations of the system to guarantee reliable and functional software.

Usability requirements

- NFR1: New users should only need to spend a maximum of 30 minutes being trained on the user interface.
- NFR2: Role-specific dashboards (Head Chef, Delivery Person) will be displayed by the app.

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- NFR3: The interface must have a consistent design and be usable on both desktop and mobile devices.
- NFR4: Important alerts (low stock, expiration) must be easily recognizable and accessible with just two touches.

Reliability requirements

- NFR5: During business hours (7 a.m. to 11 p.m.), the system must maintain 99.5% uptime.
- NFR6: Every 60 seconds, the system will automatically save inventory data to a secure database.
- NFR7: The system will resume in its most recent stored state in the event of a network or power outage.
- NFR8: Following each insert, remove, or reorder transaction, data integrity needs to be confirmed.

Performance requirements

- NFR9: The system must handle actions to add or remove items in less than two seconds.
- NFR10: Within five seconds of a triggering event, alerts and notifications must be provided.
- NFR11: Up to 20 users can use the app at once without experiencing any performance issues.
- NFR12: The database must have a maximum query response time of one second and be able to process at least 10,000 inventory records.

4. Interfaces

This section outlines how the FFSmart system connects with users, hardware, and supporting software. The aim is to give a clear picture of what each interface looks like, how it behaves, and how it supports the system's operation.

User interfaces

The system will present a set of interfaces that match the responsibilities of different user roles. The layout is simple and designed so that new users can learn it quickly. All screens follow a consistent style, with the same navigation structure, buttons, colours, and icon patterns. Alerts appear in a clear and noticeable format, and any errors are shown at the top of the screen with instructions to fix the issue.

Head

Chef

Interface

This interface provides a full overview of the current inventory, expiry conditions, and low-stock levels. The head chef can confirm automated reorders, add or update chef accounts, and review health and safety reports. The design focuses on quick access to important information so that

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operational decisions can be made without delay.

Chef

Interface

Chefs receive a simplified interface that focuses on adding or removing items and checking inventory. This keeps their workflow fast during busy periods in the kitchen. They do not have access to configuration features or reports.

Create a design

Delivery

Interface

Delivery staff have an interface designed for checking scheduled deliveries, logging received stock, and gaining access to the rear delivery entrance. The screens are kept minimal to reduce delays during stock drop-off.

Health

and

Safety

Officer

Interface

This is a read-only view that shows compliance data, expiry trends, logs, and export options. No editing tools are provided since this role only needs to review information.

System

Administrator

Interface

Administrators can view and manage user roles, system settings, backups, and activity logs. This interface shows more technical details of the system but follows the same design structure so it stays consistent with the other user views.

Hardware interfaces

The current project is delivered as a simulation, so the software does not yet communicate with physical fridge hardware. However, the design includes hardware interface preparation for future integration when the FFSmart fridge units are fully deployed.

The hardware will include sensors for temperature, item weight, barcode scanning, and door access. When integration takes place, the system will receive sensor data and update the database automatically. The delivery module will also interact with the electronic locking system for the rear door, responding to valid delivery logins.

For development and testing, a simulated hardware module is used. This module provides sample temperature readings, inventory changes, and delivery events so the system can run as if the fridge were already connected.

Software interfaces

The FFSmart system consists of several software components that work together. Each component communicates through clearly defined interfaces to keep the system organised, maintainable, and secure.

Database Interface

The system connects to a relational database that stores inventory records,

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user accounts, supplier details, logs, and order history. The application communicates with the database through a secure access layer to prevent accidental or unauthorised changes. This layer ensures that all operations use validated inputs.

Backend

Services

Core system logic, such as inventory updates, reorder creation, security checks, and notification handling, is processed by backend services. These services interact with both the user interface and the database. Keeping these services modular makes it easier to update or expand the system in future versions.

Access Control Interface

Login requests go through the authentication module, which validates user credentials and applies role-based permissions. Passwords are encrypted before being stored in the database. Access checks are performed before any restricted action is completed.

Reporting

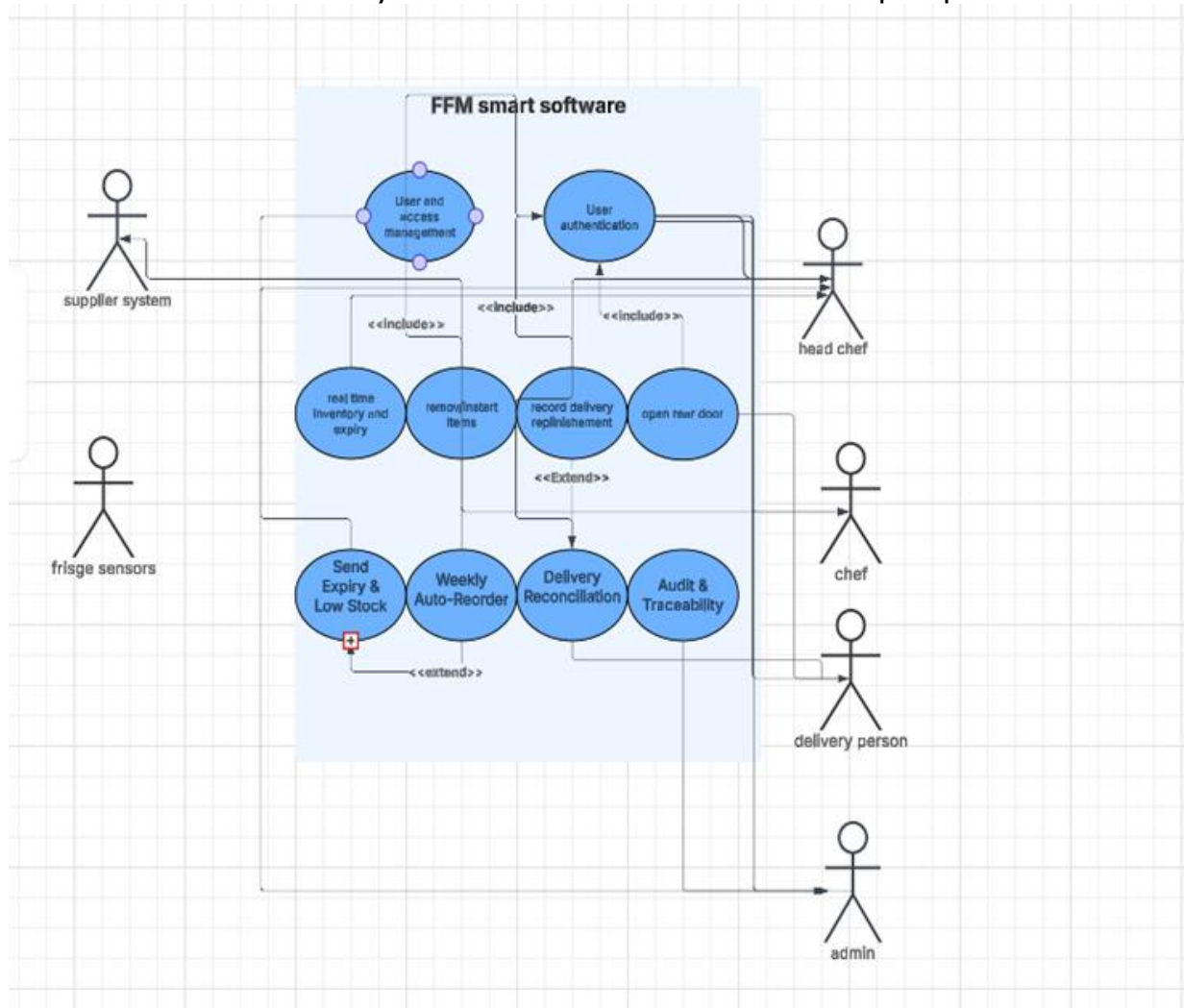
Interface

The reporting module gathers stored data from logs and inventory history. It formats this information into exportable documents for compliance checks and internal reviews. This interface is available to the Head Chef, the Health and Safety Officer, and the System Administrator.

5. Use Case Modelling

This section describes how each actor interacts with the FFM Smart Software. The diagram shows the relationships between users (Head Chef, Chef, Delivery Person, Admin), connected systems (Fridge Sensors and Supplier System), and the main functional processes of the application. Each use case

below summarises the system behaviour from the user's perspective



5.1 Use Cases

UC1 – User Authentication

Actors	Head Chef, Chef, Delivery Person, Admin
Description	Enables authorised users to securely access the system.
Preconditions	Valid user account exists in the database.
Main Flow	1. User enters credentials. 2. System verifies details. 3. Appropriate dashboard loads based on role.
Alternative Flow	Invalid credentials → error message displayed.

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Postconditions	User session initiated and recorded in the audit log.
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UC2 – User and Access Management

Actors	Admin
Description	Allows the Admin to create, update and remove user accounts and assign appropriate roles.
Preconditions	Admin is logged in.
Main Flow	Admin opens User Management → adds or edits accounts → saves changes.
Alternative Flow	If permissions invalid, operation is rejected.
Postconditions	User database updated with new access levels.

UC3 – Real-Time Inventory and Expiry Tracking

Actors	Fridge Sensors, Head Chef, Chef
Description	Tracks all items stored in the fridge and automatically updates quantity and expiry data.
Preconditions	Sensors connected and active.
Main Flow	Sensors detect quantity/expiry → system updates database → displayed in dashboard.
Alternative Flow	If sensor offline, system logs missing data.
Postconditions	Inventory table refreshed with current stock levels.

UC4 – Remove / Insert Items

Actors	Chef, Head Chef
Description	Records addition or removal of stock items from the fridge.
Preconditions	User logged in and item exists in inventory.
Main Flow	User selects item → specifies quantity change → system updates inventory.
Alternative Flow	If invalid quantity, prompt for correction.
Postconditions	Stock quantity adjusted and logged for audit.

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UC5 – Record Delivery Replenishment / Open Rear Door

Actors	Delivery Person, Head Chef
Description	Confirms deliveries and updates stock while granting controlled access to the rear delivery door.
Preconditions	Delivery Person authenticated.
Main Flow	Delivery Person requests door access → system verifies → records delivered items.
Alternative Flow	Unscheduled delivery → access denied.
Postconditions	Inventory updated; delivery event logged.

UC6 – Send Expiry and Low Stock Alerts

Actors	System (automated), Head Chef
Description	Notifies responsible staff when stock is running low or near expiry.
Preconditions	Inventory thresholds configured.
Main Flow	System analyses inventory → detects triggers → sends notification.
Alternative Flow	Notification service offline → alert queued.
Postconditions	Alert issued; event recorded.

UC7 – Weekly Auto-Reorder

Actors	System (automated), Supplier System
Description	Automatically generates and sends supplier orders for items below threshold.
Preconditions	Supplier accounts set up.
Main Flow	System compiles list → sends order request via API → awaits confirmation.
Alternative Flow	If supplier unavailable → retry scheduled.
Postconditions	Reorder logged and inventory forecast updated.

UC8 – Delivery Reconciliation / Audit & Traceability

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Actors	Admin, Health and Safety Officer
Description	Provides audit trail of inventory activity, deliveries and user actions.
Preconditions	System records maintained.
Main Flow	User selects report → system retrieves and formats data → exports to file.
Alternative Flow	If record missing, display warning.
Postconditions	Report generated for compliance.

5.2 Misuse Cases

Misuse Case 1 – Unauthorised Access Attempt

Threat	Attempts to log in using invalid credentials to view stock data.
Mitigation	Account lockout after three failed attempts and two-factor authentication.

Misuse Case 2 – Data Tampering by Chef

Threat	Chef tries to modify supplier or expiry data.
Mitigation	Role-based access control restricts editing privileges to Head Chef and Admin only.

Misuse Case 3 – Log Deletion Attempt

Threat	User attempts to remove audit logs to hide actions.
Mitigation	Logs are write-once and accessible to Admin only; backup copies stored securely.

Misuse Case 4 – Supplier System Spoofing

Threat	Malicious user sends fake supplier responses to trigger incorrect reorders.
Mitigation	All supplier requests verified via API key and encrypted communication channels.

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Misuse diagram

6. Project Plan

The project team was organised so each member had a clear role and responsibility throughout the development process. This structure helped the group divide work fairly, stay coordinated, and maintain steady progress as the system was designed and documented. Each member focused on specific areas but also supported others where needed.

The roles were distributed as follows:

Project Coordinator

Responsible for arranging meetings, keeping track of deadlines, and making sure each section of the report progressed as planned. This role also involved checking that everyone understood their tasks and resolving any uncertainties early.

Requirements Lead

Focused on gathering, structuring, and writing the functional and non-functional requirements. This ensured the group had a shared understanding of what the FFSmart system must deliver before any modelling or planning was carried out.

Design and Modelling Lead

Prepared the use case diagrams, use case descriptions, misuse cases, and supported the early system design decisions. This role ensured consistency across all modelling sections and kept the diagrams aligned with the requirements.

Technical Planning Lead

Responsible for researching appropriate technologies, drafting the project management tools section, and supporting the team in deciding how the system will be built in later development stages.

Documentation Lead

Ensured all written work flowed clearly, matched the report structure, and followed the template. This role involved proofreading, formatting, merging sections, and ensuring the final document looked consistent.

Although each member had individual responsibilities, the group reviewed key sections together. This allowed everyone to contribute ideas to major decisions such as requirements, methodology, and tool selection. The organisation helped maintain accountability and kept the workflow manageable during busier periods.

The team chose to follow an Agile-influenced approach for planning and

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preparing the FFSmart system. The project involves changing requirements, multiple user roles, and ongoing feedback from the client, so an iterative style fitted the nature of the work better than a rigid sequential method.

Agile allows the team to revise and refine parts of the system as new information becomes available. It also supports breaking the work into smaller tasks, which made it easier to track progress and adjust workloads during the project. The group held short check-ins to confirm what was completed, what needed more clarity, and what tasks would come next.

While a traditional Waterfall approach is good for projects where all requirements are fixed from the start, it is less flexible if updates or corrections are needed later. Since this system includes automation, hardware simulation, and role-based features, the team decided that having room to revise requirements and designs would lead to a more accurate final solution.

Advantages of the Agile-based approach for this project include:

- the ability to update requirements without disrupting the entire plan
- frequent communication that keeps the group aligned
- shorter cycles for reviewing work
- easier redistribution of tasks when deadlines shift

A limitation is that it requires the team to meet regularly and stay organised, otherwise tasks may fall behind. To manage this, the team combined Agile practices with some structure from Waterfall, especially in documentation and requirement gathering. This hybrid approach provided flexibility while still following the academic submission structure.

6.3. Project Management and Tools

The team will employ a collection of integrated technologies to provide efficient collaboration, visibility, and control throughout the Frost Byte smart software development lifecycle. These technologies facilitate transparent, cooperative development and uphold the principles of Agile project management.

Tool Type	Tool Name	Purpose	Justification
Progress Tracking	Jira	used to create and oversee tasks, sprints, and user stories; Kanban boards are used to visually monitor progress.	Agile sprint planning, task ownership, and integrated reporting tools for tracking progress and velocity are all made possible by these tools.
Team Communication	Microsoft Teams	During development, it is	Both lessen the need for lengthy email

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		utilized for file sharing, brief conversations, and real-time communication.	threads, offer structured communication channels, and integrate with project tools.
Document Sharing	OneDrive	central area for project information, including reports, meeting minutes, and design documents.	Version control, simultaneous editing, and accessibility from any device are all possible with cloud-based systems.
Source Control	Github	All project source code is stored using version control; pull requests, branching, and merging are supported.	An industry-standard tool that keeps track of all code changes and encourages teamwork in development.
Development Environment (IDE)	Visual Studio Code	Front-end and back-end components are primarily coded in development environments.	Support for several languages, Git integration, debugging tools, and extensive community support.
Testing Framework and Bug Reporting	Github	used to log and track defects during quality assurance, as well as automate unit and integration testing.	guarantees consistent test coverage and promotes test-driven development (TDD), while GitHub Issues centralizes fault reporting.
Diagramming and Modelling	LucidChart	used to produce architecture designs, flowcharts, and UML diagrams.	provides drag-and-drop capability, OneDrive and Jira connectivity, and simple collaboration.

Combining these tools ensures efficiency and accountability by offering complete visibility throughout the project's phases, from requirement collecting to deployment.

Meeting evidence -- screen shot

6.4. Risk assessment

The following table lists possible project hazards, assesses their impact and likelihood, and outlines mitigation techniques to lessen their consequences.

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Risk ID	Description	Likelihood	Impact	Mitigation Strategy
R1	Uncertain or changing client needs	Medium	High	Keep in constant contact with the client broker and record any modifications to the agreed-upon requirements.
R2	Task completion delays or slippage of milestones	Medium	High	Prioritize core features first and use Agile sprints with rigorous weekly reviews (MoSCoW).
R3	Uneven participation or the absence of a team member	Low	High	Maintain shared documents, cross-train employees, and assign tasks equitably.
R4	Technical challenges or problems with software integration	Medium	Medium	To test components early, use continuous integration and modular design.
R5	Data corruption or loss	Low	High	Use cloud-based storage for redundancy, frequent backups, and version control.
R6	Security flaws (data leakage, illegal access)	Low	High	Use role-based access control, encrypt important data, and adhere to safe coding standards.
R7	Software incompatibility with upcoming refrigerator hardware	Medium	Medium	Plan future integration testing and create software as a stand-alone prototype with simulated hardware interfaces.
R8	Customer discontent	Medium	Medium	Involve stakeholders in

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	caused by poor UI/UX			prototype reviews and conduct frequent usability testing.
R9	Ineffective team member communication	Low	Medium	For transparent collaboration, set up weekly check-ins and make use of shared technologies like Teams and Trello.

6.5 Task Estimation

6.5.1 Task Estimation Technique

Task durations were estimated after team discussion. Each member proposed how long their assigned work would realistically take based on its complexity. The schedule was then turned into a 10-week plan ending on 6 February 2026.

6.5.2 Gantt chart

Task Name	Task Name	Start Date	End Date	Week 1 (25 Nov)	Week 2 (02 Dec)	Week 3 (09 Dec)	Week 4 (16 Dec)	Week 5 (23 Dec)	Week 6 (30 Dec)	Week 7 (06 Jan)	Week 8 (13 Jan)	Week 9 (20 Jan)	Week 10 (27 Jan)
Requirements Gathering	Naufal	25/11/2025	06/12/2025										
System Design	jay/mujeedat	07/12/2025	15/12/2025										
Database Implementation	tyrese	16/12/2025	26/12/2025										
Backend Development	Tyrese	27/12/2025	10/01/2026										
Frontend Development	full team	05/01/2026	20/01/2026										
Integration & Testing	mujeedat	21/01/2026	28/01/2026										
Documentation	jay/naufal	29/01/2026	03/02/2026										
Final Review & Submission	full Team	04/02/2026	06/02/2026										

6.5.3 Milestones

Requirements completed	06/12/2025
System design completed	15/12/2025
Database backend completed	10/01/2026
Ui integration completed	20/01/2026
Testing completed	29/01/2026
Final submission	06/02/2026

Milestone – diamond in chart

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improve Gantt chart overall

6.6 Task Estimation

The following procedures have been established to guide the team in task management, meeting deadlines and decision making. These procedures are designed to help us stay organised, maintain accountability and have good communication throughout the project process. Any hinderances or issues will be dealt with efficiently and effectively. The procedures are as follows:

Task deadline not met:

- Inform and explain to the group as early as possible why there's delay to task completion. In which discussion can be had to decide whether task deadline should be extended, parts should be reassigned or if any members can provide additional support. Changes to deadlines will be recorded in project tracker.
- Any issues with deliverables being delivered by deadline date set, should be brought up to the group at least 48 hours beforehand. Repeated missing of deadlines will be taken note off.

Lack of Participation and Attendance

- Attendance to scheduled meeting is expected by all members of the team unless a valid response is given prior to the meeting.
- Repeated attendance or lack of participation should result in firstly an informal reminder of the expectation. If failure to attend persist then note should be taken to maintain transparency of communication between the group and members who fail to attend. Task may be reassigned depending on necessity and module tutor may be informed if required.

Selection of Technologies

- Team members will research and present options for possible appropriate technologies and justify the uses of them, to where an evaluation; during scheduled meetings, will lead to confirmation of technologies used based on the plausibility against the criteria.
- Consensus should be made democratically and changes to technology should only be changed if team agrees
- Factors that should be considered:
 - Compatibility with project requirements
 - Ease of use or learning
 - Performance and scalability
 - Familiarity amongst group members

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Contribution table

Name	Primary Role(s)	Areas worked on	Contribution rating (0-4)
Naufal	Project Coordinator, Design and Modelling Lead	Section 5, section 6	4
Mujeedat	Documentation Lead	Section 3, section 6	4
Tyrese	Requirements Lead	Section 1,2, 6	4
Jay	Technical Planning Lead	Section 4, section 6	4

Signed by:

Name	Signed
Naufal	<i>Naufal</i>
Mujeedat	<i>Mujeedat</i>
Tyrese	<i>Tyrese</i>
Jay	<i>Jay</i>

Contribution ratings (0-4)

Contribution rating	Meaning	Description
4	Full contribution	The student has contributed generally well and/or has provided a satisfactory amount of work throughout the project. A typical example would be a student that attended all or many meetings, and acted professionally throughout (e.g. was reliable enough to let people know when not attending meetings etc.) and contributed to large parts of developing the program and/or report.
3	Mid contribution	The student has done a medium amount of work throughout the project. A typical example would be a student that may not have attended/helped at all the meetings they should have attended, but was generally available and helpful. They may have taken on simpler parts of the project that did not demand a large portion of effort/diligence.

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2	low contribution	The student has done the bare minimum. A typical example would be a student that has missed many meetings, not contributed when they did join, and knew very little about what was going on/what was required. Another example would be a student that turns up at the last minute (e.g. a few days before the deadline) to help and delivers something reasonable.
1	Very low contribution	The student has done very close to nothing overall. A typical example would be a student the group rarely sees over the entire time and has provided nothing substantial for the coursework. They may turn up at the last minute and provide something. They were generally a poor communicator and/or would produce very little when asked.
0	No contribution	The student has done absolutely nothing overall. A typical example would be a student the group has never seen over the entire time and has provided nothing for the coursework.